

ESE6180 Project Proposal: (tentative title) From Evaluation to Improvement: Boundary-Guided Training of Closed-Loop Controllers

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Abstract

A frozen controller’s failures across operating conditions form a map of where it works. We propose to investigate whether training near an estimated performance boundary expands that region while limiting regression where the controller was already acceptable. We certify the region of a frozen policy, feed that estimate into the choice of training conditions, and re-certify on fresh data. Monotonic improvement is the ambitious target, not an assumption.

1. Motivation, Problem Formulation, and Feedback Loops

Evaluating a frozen controller across operating conditions maps where it works, not just how well. On CartPole with the pole-angle cutoff relaxed to 90° , call an episode a *recovery* if it survives 500 steps with final-100-step RMS pole angle at most 5° . An adaptive sweep of initial pole angles and half-lengths using [this experiment repository](#) gives Figure 1: quantized LQR and PPO have recovery regions differing in shape, not only size, and survival and recovery disagree. We ask whether evaluating that boundary can guide training that *expands* the acceptable region while limiting degradation elsewhere, which signal—threshold proximity, intermediate difficulty, uncertainty, or improvability—marks a useful training condition, and what assumptions permit a guarantee. Informative *evaluation* conditions need not be informative *training* conditions.

At iteration k , freeze π_k . For a condition $z \in \mathcal{Z}$, horizon H , and declared episode randomness $\xi \sim P_\xi(\cdot | z)$, let $p_k(z) = \Pr_\xi[\tau_{\pi_k}(z, \xi) \text{ fails recovery}]$; under tolerance α the acceptable region is $S_k = \{z : p_k(z) \leq \alpha\}$. With a measure μ fixed in advance, $V_k = \mu(S_k)$, $g_k = \mu(S_{k+1} \setminus S_k)$, and $\ell_k = \mu(S_k \setminus S_{k+1})$ satisfy $V_{k+1} - V_k = g_k - \ell_k$. Three loops nest—**control**: observation $\rightarrow$ action $\rightarrow$ next observation; **evaluation**: condition selection $\rightarrow$ frozen rollout $\rightarrow$ updated evidence $\rightarrow$ next condition; and **improvement**, studied here:

$$\pi_k \longrightarrow \hat{p}_k, \hat{S}_k \longrightarrow q_k = (1 - \lambda)q_{\text{base}} + \lambda q_{\text{target},k} \longrightarrow \text{Train}(\pi_k, q_k) \longrightarrow \pi_{k+1}.$$

Closing it moves the target from p_k to p_{k+1} ; policies change only in a separate training phase, so each $\hat{p}_k$ describes one closed-loop system. What is distinctively L4DC is that the estimated quantity is a closed-loop property of a learned controller over initial conditions and plant parameters, not a reward, and that the object to certify and grow is a region, not a scalar.

2. Related Work

Evaluation. [Gotovos et al. \(2013\)](#) and [Letham et al. \(2022\)](#) develop adaptive level-set estimation, the latter for Bernoulli observations like ours; [Howard et al. \(2021\)](#) supply time-uniform confidence sequences, so adaptive sampling still certifies. **Training-condition selection.** Reverse curriculum generation ([Florensa et al., 2017](#)) adapts start states to current performance, prioritized level replay

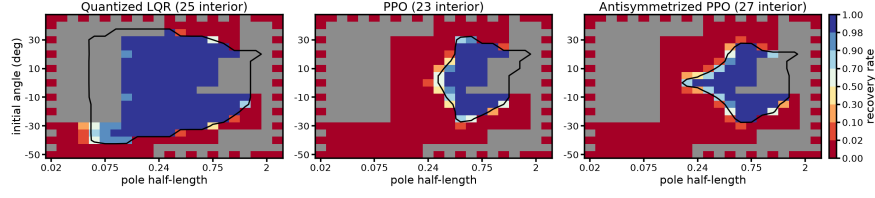


Figure 1: Preliminary recovery rate for three *frozen* controllers on the angle/length lattice in mesh-rank coordinates (248 of 440 slots sampled, gray unsampled; black line is the 0.5 contour). Over 92% of cells are exactly 0 or 1: the interior cells are the whole boundary.

(Jiang et al., 2021) ranks configurations by learning potential, and sampling for learnability (Rutherford et al., 2024) targets mixed-success conditions, so boundary-guided sampling is not itself novel; our contribution is the controlled comparison under a fixed, certified protocol. **Control guarantees.** Berkenkamp et al. (2017) link model-based learning to safe-region expansion via Lyapunov analysis; their stability certificates differ from our finite-horizon recovery probabilities.

3. Goals (tentative)

Stage 1 (course foundation). Build the evaluator: certify a condition only when a confidence sequence puts its failure rate strictly on one side of α , and split $\sum_k \delta_k \leq \delta$ so all certified sets hold with probability $1 - \delta$ (proved by Hoeffding and a union bound). For $x_{j+1} = (A - BK)x_j$ and $J_{H,K}(x) = x^\top P_H(K)x$ on $\|x\|_2 \leq r$, the update bound $|J_{H,K'}(x) - J_{H,K}(x)| \leq r^2 \|P_H(K') - P_H(K)\|_2 =: b$ confines all regression of the sublevel set $\{J_{H,K} \leq h\}$ to the band $h - b < J_{H,K} \leq h$. These elementary results do not transfer automatically to the nonlinear learned system.

Stage 2 (main research). From one frozen PPO checkpoint, branch into matched training runs under three selection rules—uniform, boundary-guided over a band at the estimated boundary, and intermediate-difficulty scored by $\hat{p}_k(1 - \hat{p}_k)$ —holding q_{base} , λ , the objective, and the optimizer fixed. Starting with initial-angle variation, then the angle-length domain of Figure 1, we report g_k , ℓ_k , net coverage, maps of *where* change occurs, seed variation, and total cost. Each updated policy is re-certified on fresh data, separating certificate growth from retraining. A negative result is a real outcome.

Stage 3 (ambitious target). Bound performance drift by update size, horizon, and closed-loop amplification; construct examples where improving selected conditions degrades others; and find a restricted controller class and update admitting a sufficient condition for expansion. Both the continuous-cost sublevel set and the level set S_k are candidates. **Open decisions.** Whether repeated rounds are required or a single $\pi_0 \rightarrow \pi_1$ update suffices, and which level-set notion is primary, stay open; α , δ , budgets, seeds, reward, and mixture weights will be fixed and recorded beforehand.

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